

Decent Coaching Classes

IX CBSE

Mind Your Math - II

Answer Key: Test - Chapters 2, 4, 5 (up to 5.3) and 7

SECTION A — MCQs

1. (C) 2
2. (B) $a^2 - 2ab + b^2$
3. (B) diameter
4. (C) infinite
5. (C) 1

SECTION B — Very Short Answer

6. Slope (a) = 3, y-intercept (b) = 5
7. $(2x + 3y)^2 = 4x^2 + 12xy + 9y^2$
8. Chord: a line segment joining any two points on a circle. Diameter: a chord that passes through the centre of the circle (the longest chord).
9. Sample space $S = \{HH, HT, TH, TT\}$; sample size $n(S) = 4$

SECTION C — Short Answer

10. Let the smaller number = x, larger = x + 12. Sum: $x + (x + 12) = 48 \Rightarrow 2x = 36 \Rightarrow x = 18$. The two numbers are 18 and 30.
11. $(204)^2 = (200 + 4)^2 = 200^2 + 2 \times 200 \times 4 + 4^2 = 40000 + 1600 + 16 = 41616$
12. Since $6^2 + 8^2 = 36 + 64 = 100 = 10^2$, $\triangle ABC$ is a right-angled triangle (right angle at the vertex opposite the 10 cm side, i.e., at B). By the property of the circumcentre of a right-angled triangle, the circumcentre lies at the midpoint of the hypotenuse — that is, it lies ON the triangle (on side CA), not strictly inside or outside it.
13. Total balls = 5 + 3 + 2 = 10. (i) $P(\text{red}) = 5/10 = 1/2$ (ii) $P(\text{not green}) = 1 - 2/10 = 8/10 = 4/5$ (iii) $P(\text{blue or red}) = 8/10 = 4/5$

SECTION D — Long Answer

14. Given $x + y + z = 12$, $xyz = 28$, $x^2 + y^2 + z^2 = 66$.
(i) $(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx) \Rightarrow 144 = 66 + 2(xy + yz + zx) \Rightarrow xy + yz + zx = 39$
(ii) $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx) = 12 \times (66 - 39) = 12 \times 27 = 324$
 $\therefore x^3 + y^3 + z^3 = 324 + 3(28) = 324 + 84 = 408$